Hybridization and Sigma and Pi Bonds

sp^3, sp^2, sp and bond types

Name _____ Class _____ Date _____

RULES

Electron domains = bonded atoms + lone pairs (a double or triple bond counts once)

4 → sp^3 (109.5°) · 3 → sp^2 (120°) · 2 → sp (180°)

Single = 1σ · double = 1σ + 1π · triple = 1σ + 2π

Draw a Lewis structure first when you need one.

A. Hybridization

1. Explain what is meant by hybridization, and why it is needed to explain the shape of methane.

2. Give the hybridization of the named atom and the ideal bond angle around it: (a) C in CCl_4 (b) B in BF_3 (c) C in HCN (d) N in NH_3 (e) C in $H_2C{=}O$ (f) Be in $BeCl_2$

B. Sigma and pi bonds

3. Describe the difference between a σ bond and a π bond.

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4. Count the σ and π bonds in: (a) ethane, C_2H_6 (b) ethene, C_2H_4 (c) N_2 (d) propene, $CH_2{=}CH{-}CH_3$ (e) CO_2

C. Carbon in organic molecules

5. For propyne, $HC{\equiv}C{-}CH_3$, give the hybridization of each carbon atom (from left to right) and the bond angles around it.

6. Explain why there are two different forms (isomers) of 1,2-dichloroethene, $ClCH{=}CHCl$, but only one form of 1,2-dichloroethane, $ClCH_2{-}CH_2Cl$.

D. Lone pairs

7. Water and carbon dioxide both have a central atom bonded to two others. Explain why the oxygen in water is sp^3 but the carbon in CO_2 is sp, and how this affects their shapes.

E. Challenge

8. Proteins contain the amide group $-C(=O)-NH^-$. In methanamide, $H-C(=O)-NH_2$: (a) give the hybridization of C and O (b) count the σ and π bonds (c) predict the O=C–N bond angle.

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